

SOFTWARE REQUIREMENTS SPECIFICATION

TeaRoutePay

Tea Route Collection & Payment System

Document ID	TCPS-SRS-2026-v1.0
Date	May 29, 2026
Repository	seng31242-tea-route-pay-2026 / tea-collection-payment-system
Status	FINAL
Prepared by	Group 09 – SENG 31242 (2026)

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1. Introduction

1.1 Purpose and Scope

This Software Requirements Specification (SRS) defines the complete functional and non-functional requirements for TeaRoutePay – a multi-platform tea route collection and payment management system. This system replaces the current paper-based field process and legacy Windows desktop application. The system modernizes tea leaf collection recording, farmer profile management, deduction processing, monthly payment calculation, bank transfer disbursement, and SMS farmer notification.

This document governs design, implementation, system testing, acceptance testing, and future maintenance.

1.2 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
TCPS	Tea Collection & Payment System (alternate short form used in IDs)
SRS	Software Requirements Specification
UUID	Universally Unique Identifier
FR	Functional Requirement (numbered FR-01, FR-02 ...)
NFR	Non-Functional Requirement
UC	Use Case
SSD	System Sequence Diagram
RBAC	Role-Based Access Control
API	Application Programming Interface
REST	Representational State Transfer
LAN	Local Area Network
QR	Quick Response code – used for farmer identification
SMS	Short Message Service
LKR	Sri Lankan Rupee
SQLite	Embedded relational database for offline mobile / office local storage
TLS	Transport Layer Security
JWT	JSON Web Token – used for session authentication
ORM	Object-Relational Mapping framework
CI/CD	Continuous Integration / Continuous Delivery
BR	Business Rule

1.3 References

- SENG 31242 System Design Project Course Guideline & Industrial Standards
- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
- GitHub Repository: <https://github.com/seng31242-tea-route-pay-2026/tea-collection-payment-system>

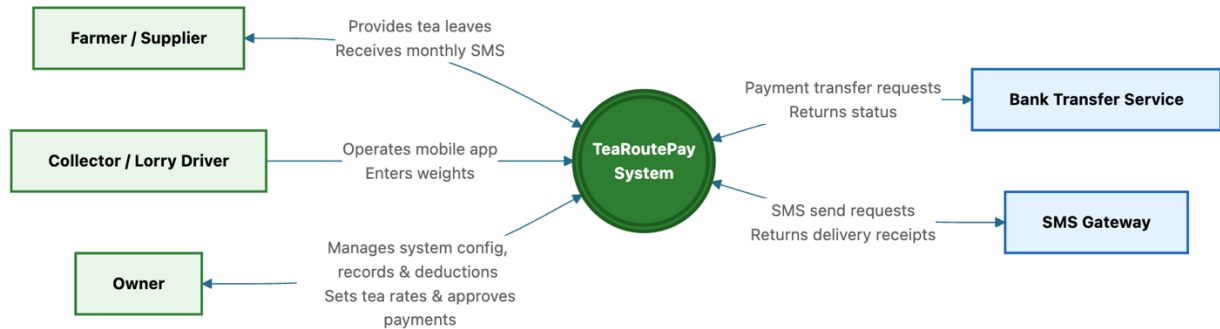
1.4 Overview of Document

1. Section 1 – Introduction: purpose, scope, definitions, references.
2. Section 2 – Overall Description: product perspective, user classes, operating environment, constraints, assumptions.
3. Section 3 – System Analysis: fact-gathering techniques, existing system analysis, use-case diagrams, use-case descriptions, activity diagrams, and system sequence diagrams.
4. Section 4 – Functional Requirements: numbered FR list (FR-01 ... FR-53) with ID, title, description, priority, source, and verification method.
5. Section 5 – Non-Functional Requirements: performance, security, usability, reliability, scalability, maintainability.
6. Section 6 – Alternative Solutions & Feasibility Study: two alternatives analyzed, four-dimension feasibility, justified recommendation.

2. Overall Description

2.1 Product Perspective (Context Diagram)

TeaRoutePay is a greenfield replacement for a legacy Windows desktop application and manual paper-based route books. The diagram below illustrates the system boundary and all external entities.



Context boundary – external entities that interact with TeaRoutePay:

External Entity	Interaction	Direction
Farmer / Supplier	Provides tea leaves; receives monthly SMS payment notification	Two-way
Collector / Lorry Driver	Operates mobile app during field route; enters weights	Input
Office Staff / Owner	Manages farmer records, deductions, reports, payment calculation Sets tea rate, approves payment run, views dashboard and audit log Manages users, roles, sync mode, system configuration	Both
Bank Transfer Service (ext.)	Receives approved payment transfer request; returns status	Both
SMS Gateway (ext.)	Receives SMS send requests; returns delivery receipts	Both

2.2 User Classes and Characteristics (Personas)

User Class	Persona	Technical Level	Primary Interface	Key Responsibilities
Office Staff	Nimal, 52, business proprietor and office administrator. Comfortable with Windows software.	Medium	Desktop App	Set monthly rate, approve payment, view reports and audit log. Farmer registration, receipt printing
Collector	Suresh, 28, lorry driver. Uses a smartphone. Follows a route daily.	Low	Mobile App	QR scan, weight entry, data sync at end of route

2.3 Operating Environment

1. **Mobile Application:** Android 8.0 (API 26) + or iOS 13.0+. Min 2 GB RAM. Device camera required for QR scanning.
2. **Office Desktop Application:** Windows 10 (64-bit) or Windows 11. Min 4 GB RAM, 2 GHz CPU, 10 GB free disk.
3. **Backend / API Server:** Linux (Ubuntu 20.04 LTS+). Min 4 GB RAM, 2 vCPUs.
4. **Local Database (Mobile):** SQLite (embedded, no separate install).
5. **Local Database (Office):** PostgreSQL 13+.
6. **Cloud Database:** PostgreSQL 13+ on cloud host (when cloud sync mode is selected).
7. **Network:** No network required for field collection. LAN (802.11n Wi-Fi / 100 Mbps Ethernet) for local sync. Internet (min 2 Mbps) for cloud sync.

2.4 Design and Implementation Constraints

1. **Offline-First Resilience:** The mobile application must be fully functional offline during field collection with absolutely zero network dependency.
2. **Flexible Synchronization:** Both LAN sync (local network) and cloud sync (internet) modes must be supported. The active mode must be selectable via application configuration or UI toggle, requiring no code changes to switch.
3. **Financial Accuracy:** All monetary calculations and storage must use fixed-point arithmetic (or integer representation of cents/lowest denomination) to guarantee exact LKR accuracy and strictly prohibit the use of floating-point arithmetic to prevent rounding errors.

4. **Data Immutability & Auditing:** Approved payment records must be strictly immutable, no modifications or deletions are permitted after Factory Owner approval. Furthermore, the system must maintain an immutable, append-only audit log tracking all changes to financial records and system configurations.
5. **Data Security (In Transit & At Rest):** Farmer personal and financial data must be encrypted in transit using TLS 1.2 or higher. Data stored locally on mobile devices or offline desktop clients must be encrypted at rest (e.g., utilizing SQLCipher for embedded SQLite databases).
6. **Secrets Management:** Bank API credentials, SMS API keys, database passwords, and other sensitive configurations must be managed via a secure secrets manager or environment variables. They must never be hardcoded into the source code repository.
7. **Hardware Interoperability:** QR codes generated by the system must use standard formats (e.g., plain text IDs or standard JSON) to ensure they are scannable by standard device cameras without the need for proprietary or third-party decoding software.

2.5 Assumptions and Dependencies

1. All farmers are pre-registered in the system before any collection route is executed.
2. Unique QR code cards distributed to all registered farmers.
3. The business operates on a monthly payment cycle with one consolidated payment per farmer per month.
4. A LAN is available at the office premises when local sync mode is used.
5. Internet connectivity (via mobile data in the field or Wi-Fi at the office) is available when cloud sync mode is utilized. The mobile application can sync directly to the cloud server from any location with an active internet connection.
6. External SMS and bank transfer service providers supply documented REST APIs and test environments.
7. Tea rates are updated by the Owner before payment calculation is run.
8. Device clocks are synchronized to within 1 minute to support accurate conflict resolution.

3. System Analysis

3.1 Fact-Gathering Techniques Used

The following methods were used to collect, understand, and confirm the requirements for the system:

Technique	Description	Artefacts Produced
Client Interviews	Structured interviews with the business owner to understand current workflows, pain points, and desired outcomes. Three interview sessions conducted.	Interview transcripts, pain-point list, initial requirement seeds
On-Site Observation	Direct observation of a lorry collection route and the office payment calculation process to understand the real-world operational context.	Process notes, route timing data, paper form samples
Document Analysis	Review of existing paper route books, the legacy desktop application screenshots, and current bank transfer forms.	Legacy system analysis, data entity list
Questionnaire	Questionnaire distributed to three collectors to capture mobile usage expectations and connectivity conditions in the field.	Usability constraints, NFR inputs

3.2 Existing System Analysis

3.2.1 Current System Overview

The current process relies on two manual components:

- (1) paper route books carried by collectors in the lorry
- (2) Legacy Windows desktop application used by office staff to re-enter field data and calculate monthly payments.

3.2.2 Pain Points Identified

#	Pain Point	Impact	Proposed Resolution
P-01	Field data recorded on paper, manually re-entered in office → transcription errors	Payment inaccuracies, farmer disputes	Offline mobile app with direct digital entry
P-02	Legacy desktop has no network capability → no real-time data sharing	Delays in payment processing	Backend API with LAN/cloud sync
P-03	Payment calculation is manual and slow (2–3 days per month)	Staff overtime, delayed farmer payments	Automated calculation engine
P-04	No audit trail for financial changes	Cannot investigate disputes	Immutable audit log for all financial records
P-05	Farmers do not receive notifications when the collected tea is entered into the office system and payments are made.	Farmers make unnecessary office trips	Automated SMS on payment confirmation
P-06	Deduction tracking is done manually	Errors in net payment	Integrated deduction management module
P-07	Paper QR-less identification	Collectors must write their name and supplier number with each collection round.	QR code-based farmer identification

3.3 Use-Case Diagram (System Boundary)

Use-case diagrams are presented per actor below. A combined system overview diagram should also be created showing all actors, all use cases, and include/extend relationships within the system boundary of TeaRoutePay.

3.3.2 Tea Collector (Mobile App)

- UC01: Log in to mobile app
- UC02: View assigned route
- UC03: Scan farmer QR code
 - UC03a: Record leaf weight
- UC04: Record leaf weight
 - UC04a: Save record offline
- UC05: Item distribution and record keeping
 - UC05a: Save record offline
- UC06: Issue advance payment and record keeping
 - UC06a: Save record offline
- UC07: Sync collection data
 - UC07a: Local LAN sync
 - UC07b: Cloud sync
- UC09: Register new farmer (*Actor: Tea Collector*)
 - UC09a: Generate or add QR code

3.3.3 Owner / Office Staff (Desktop App)

- UC08: Log in to desktop app
- UC09: Register new farmer (*Actor: Owner / Office Staff*)
 - UC09a: Generate or add QR code
- UC10: Manage farmer profiles
- UC11: Manage deductions
 - UC11a: Add loan deduction
 - UC11b: Add fertilizer deduction
 - UC11c: Other deductions
- UC12: Set monthly tea rate
- UC13: Calculate monthly payment
 - UC13a: Apply all deductions
- UC14: View payment summary
- UC15: Approve payment run
- UC16: Process bank transfer – (*interacts with Bank API*)
- UC17: Send SMS notification – (*interacts with SMS API*)
- UC18: Generate reports
- UC19: Manage user accounts
- UC20: Manage collection routes

3.3.3 Farmer (External / Physical Actor)

- UC21: Receive QR code card
- UC22: Provide leaves for weighing
- UC23: Receive monthly payment – (*via Bank API*)
- UC24: Receive fertilizer and other items
- UC25: Receive SMS – (*via SMS API*)

3.4 Use-Case Descriptions

3.4.1 UC - 01 – Log in to mobile app

Use Case ID	UC-01
Use Case Name	Log in to mobile app
Actor(s)	Collector
Preconditions	The mobile app is installed, and the Collector has an active account.
Postconditions – Success	Collector is authenticated and the main dashboard is displayed.
Postconditions – Failure	Access is denied; error message displayed.
Main Flow	1. Collector opens the app. 2. Collector enters Username/Phone and Password/PIN. 3. Collector taps "Log In". 4. System validates credentials against the database (local cache if offline). 5. System grants access and loads the dashboard.
Alternative Flow 1	At step 4: Invalid credentials. System displays "Incorrect username or password". Collector retries.
Business Rules	BR-01: If logging in offline, the system must use a securely cached token from the last successful online login.

3.4.2 UC - 02 – View assigned route

Use Case ID	UC-02
Use Case Name	View assigned route
Actor(s)	Collector
Preconditions	Collector is logged in and data has been synced.
Postconditions – Success	The route details and list of farmers for the day are displayed.
Postconditions – Failure	No route is displayed; error or "No Route Assigned" message shown.
Main Flow	1. Collector selects "My Route" from the dashboard. 2. System retrieves the daily route assigned to the Collector. 3. System displays the route name, geographical area, and the list of farmers to visit in order.
Alternative Flow 1	At step 2: No route. System finds no route assigned for the current date. Collector must contact Office Staff to assign a route and re-sync or need to check your internet connection
Business Rules	BR-02: Routes must default to displaying only farmers actively assigned to the logged-in Collector for the current calendar day.

3.4.3 UC - 03 – Scan Farmer QR Code

Use Case ID	UC-03
Use Case Name	Scan Farmer QR Code
Actor(s)	Collector
Preconditions	Collector is logged in and has selected a collection route. Farmer has a printed QR code. Device camera is functional.
Postconditions – Success	Farmer is identified and the weight entry screen is displayed.
Postconditions – Failure	Farmer is not identified; error is displayed. Collector may attempt manual selection.
Main Flow	1. Collector taps 'Scan QR' on the collection screen. 2. System activates the device camera in QR scan mode. 3. Collector positions the device over the farmer's QR code. 4. System decodes the QR code and retrieves the farmer ID. 5. System looks up the farmer record in the local database. 6. System displays the farmer's name and the weight entry form. 7. Collector visually confirms the farmer's name shown on screen.
Alternative Flow 1	At step 4: QR code is unreadable (damaged or poor lighting). System shows 'QR unreadable' message. Collector retries or selects farmer manually by name/ID.
Alternative Flow 2	At step 5: Farmer ID not found in local database. System shows 'Farmer not found'. Collector contacts Office Staff to verify registration and re-sync data.
Business Rules	BR-03: The QR code shall be generated using the farmer's NIC and a unique system identifier, with no personal data stored in plain text. BR-04: Manual fallback requires confirmation tap before proceeding to weight entry.

3.4.4 UC - 04 – Record leaf weight

Use Case ID	UC-04
Use Case Name	Record leaf weight
Actor(s)	Collector
Preconditions	Farmer has been identified via UC-03 or manual selection. A collection route session is active.
Postconditions – Success	Weight entry is saved to mobile local database with status 'Pending Sync'. Running session total is updated.
Postconditions – Failure	No record is saved. Collector is shown a validation error and must re-enter a valid weight.
Main Flow	1. System displays the weight entry form showing farmer name, route, date, and current time. 2. Collector enters the tea leaf weight in kilograms using the numeric keypad. 3. System automatically applies standard bag / water deductions (if configured) to calculate net weight. 4. Collector taps 'Save Entry'. 5. System validates the entry (ensures it is not empty or an impossible number). 6. [UC-04a] System writes the collection record to the mobile local database with status 'Pending Sync'. 7. System displays a success confirmation and returns to the scanner/route screen. 8. System returns to the collection home screen ready for the next farmer.
Alternative Flow 1	At step 5: Weight is zero, negative, or non-numeric. System highlights the field in red and displays a validation error message. Collector must correct the value before saving.
Alternative Flow 2	At Step 6: Storage Error – If device storage is full, the system displays "Device storage full." The collector must free device storage or sync pending data before continuing.
Business Rules	BR-05: Maximum weight per entry is configurable by the Administrator (default 500 kg). BR-06: A duplicate entry for the same farmer on the same date requires explicit Collector confirmation.

3.4.5 UC - 05 – Item distribution and record keeping

Use Case ID	UC-05
Use Case Name	Item distribution and record keeping (Includes UC-05a: Save record offline)
Actor(s)	Collector
Preconditions	Collector is logged in. Farmer is identified.
Postconditions – Success	Distributed item (e.g., fertilizer) is recorded against the farmer and saved locally.
Postconditions – Failure	Record is not saved.
Main Flow	1. Collector selects "Distribute Item". 2. System displays a list of available inventory (e.g., Fertilizer Bags). 3. Collector enters the quantity given to the farmer. 4. System calculates total cost (to be deducted later). 5. Collector taps "Confirm". 6. [UC-05a] System saves the distribution record locally.
Alternative Flow 1	At step 3: Exceeds limits. If the requested quantity exceeds the farmer's limit or truck inventory, system alerts the Collector.
Business Rules	BR-07: Item cost is fixed based on current inventory rates and cannot be overridden by the Collector.

3.4.6 UC - 06 – Issue Advance Payment and Record Keeping

Use Case ID	UC-06
Use Case Name	Issue Advance Payment and Record Keeping (Includes UC-06a: Save record offline)
Actor(s)	Collector
Preconditions	Collector is logged in and has a cash float. Farmer is successfully identified (via QR or manual). System has access to the farmer's accumulated collection data for the current month.
Postconditions – Success	Advance payment is successfully handed out, recorded against the farmer's profile, securely saved offline, and flagged for future deduction.
Postconditions – Failure	Advance payment is denied (e.g., exceeds allowable limit) or canceled; no record is saved.
Main Flow	1. Collector selects the "Issue Advance" option for the identified farmer. 2. System retrieves and displays the farmer's accumulated leaf weight for the current period. 3. System calculates and displays the maximum allowable advance based on the recorded weight. 4. Collector enters the requested advance cash amount. 5. System validates that the requested amount is less than or equal to the maximum allowable limit. 6. Collector physically hands the cash to the farmer and taps "Confirm Payment". 7. [UC-06a] System saves the advance transaction locally to the device's offline database with a timestamp. 8. System displays a success confirmation and updates the farmer's local ledger.
Alternative Flow 1	At step 5: Amount exceeds limit. If the entered amount exceeds the calculated allowable limit, the system blocks the entry and displays an error message. The Collector must enter a lower amount or obtain a special managerial override code to proceed.
Alternative Flow 2	At step 2: Insufficient collection history. If the farmer has zero or near-zero recorded leaf weight for the current period, the system displays "Insufficient collection for advance." The Collector must cancel the transaction.
Business Rules	BR-08: The maximum allowable advance must be calculated as a set ratio/percentage of the estimated value of the farmer's currently accumulated but unpaid tea leaves (e.g., 50% of current accumulated value). BR-09: Offline advance records act as financial receipts; they cannot be modified locally once confirmed and must be strictly synced to the desktop app.

3.4.7 UC - 07 – Synchronize Collection Data

Use Case ID	UC-07
Use Case Name	Sync Collection Data (Extends UC-07a: LAN Sync, UC-07b: Cloud Sync)
Actor(s)	Collector
Preconditions	Collector has one or more 'Pending Sync' records. Device is connected to the office LAN (local mode) or Internet (cloud mode).
Postconditions – Success	All pending records transferred and marked 'Synced'. Sync count and timestamp displayed.
Postconditions – Failure	Sync fails/aborts. Local data remains safely stored on the device for a future attempt.
Main Flow	1. Collector taps "Sync Data" on the mobile app. 2. System detects network environment (Office LAN or External Cloud). System authenticates the collector with the central server. 3. System pushes all unsynced data batches to the server. 4. Server validates and ingests the data, returning a "Success" token for those records. 5. System pulls updated data (new farmers registered by office, new monthly rates, etc.) from the server. 6. System marks uploaded local records as "Synced" and updates the "Last Sync" timestamp.
Alternative Flow 1	At step 4/5: Network drops. If the connection drops mid-sync, the system aborts. Unconfirmed records remain marked as "Unsynced" to prevent data loss.
Alternative Flow 2	At step 5: Data conflict. If a record conflicts with server data, the server accepts the record but flags it for manual review by Office Staff.
Business Rules	BR-10: Sync operations are idempotent – repeating a sync must not create duplicate records. BR-11: System retries failed sync up to 3 times with exponential back-off before notifying the user.

3.4.8 UC - 08 – Log in to desktop app

Use Case ID	UC-09
Use Case Name	Log in to desktop app
Actor(s)	Owner / Office Staff
Preconditions	Application is installed/accessible; user has credentials.
Postconditions – Success	User accesses the admin dashboard.
Postconditions – Failure	Access denied.
Main Flow	1. User enters Email and Password on the login screen. 2. User clicks "Login". 3. System authenticates credentials with the master database. 4. System loads specific dashboard.
Alternative Flow 1	At step 3: Account locked. If too many failed attempts occur, system temporarily locks the account.
Alternative Flow 2	At step 3: Missing Fields. If required fields are empty, the system disables the "Save" button and highlights the missing fields in red.
Business Rules	BR-12: The system shall restrict access to authorized users only through secure authentication.

3.4.9 UC - 09 – Register Farmer

Use Case ID	UC-09
Use Case Name	Register New Farmer (Includes UC-09a: Generate or add QR code)
Actor(s)	Tea Collector (Mobile), Owner / Office Staff (Desktop)
Preconditions	Actor is logged into the system and has "Registration" permissions.
Postconditions – Success	Farmer profile is created in the database and a QR code is generated/assigned.
Postconditions – Failure	Farmer is not registered (e.g., due to local duplicate details or missing required fields).
Main Flow	1. Actor selects "Register New Farmer" from the main menu. 2. Actor enters farmer details (Name, Address, Contact Number, NIC, Route ID, Bank Details). 3. System validates that required fields are filled and checks for duplicates (against the local database if offline, or full database if online). 4. System creates a new farmer profile with a unique Farmer ID (UUID). 5. [UC-09a] System automatically generates a printable QR code mapping to the UUID, OR Actor scans an existing pre-printed generic QR card to link it. 6. Actor taps "Save". 7. System saves the profile (syncs immediately if online, saves locally if offline).
Alternative Flow 1	At step 3: Local Duplicate found. If a farmer with the same phone number or exact name already exists in the local database/cache, the system flags a warning. Actor can choose to cancel or "Proceed anyway" (creates a new profile).
Alternative Flow 2	At step 3: Missing Fields. If required fields are empty, the system disables the "Save" button and highlights the missing fields in red.
Business Rules	BR-13: A QR code/UUID must be strictly 1-to-1 with a farmer profile. BR-14: Mobile registrations done offline must be flagged as "Pending Sync" and pushed to the master database during the next connection. BR-15: Offline Sync Conflict Resolution: If a mobile registration is synced later and the central server detects a global duplicate (e.g., phone number already exists on another route), the server accepts the record provisionally but flags it and triggers an immediate dashboard alert for the Office Staff to manually review and merge/resolve.

3.4.10 UC - 10 – Manage farmer profiles

Use Case ID	UC-10
Use Case Name	Manage farmer profiles
Actor(s)	Owner / Office Staff
Preconditions	User is logged in with Admin/Staff permissions.
Postconditions – Success	Farmer profile is updated, deactivated, or viewed.
Postconditions – Failure	Updates are not saved.
Main Flow	1. User searches for a farmer by ID or Name. 2. System retrieves and displays the profile. 3. User edits details (e.g., address, bank account info). 4. User clicks "Save Updates". 5. System validates data and updates the master database.
Alternative Flow 1	At step 5: Invalid Bank Info. If bank format is incorrect, system highlights the field and prevents saving.
Business Rules	BR-16: Deactivating a farmer does not delete their historical records; it only prevents them from appearing in active daily routes.

3.4.11 UC - 11 – Manage Deductions

Use Case ID	UC-11
Use Case Name	Manage deductions (Extends Add loan, add manual deduction, Edit synced mobile deduction)
Actor(s)	Owner / Office Staff
Preconditions	User is logged in. Mobile data (advances/items) has been synced to the database, OR staff needs to schedule a new manual deduction.
Postconditions – Success	A new deduction is scheduled, or an existing synced deduction is successfully corrected and saved to the farmer's ledger.
Postconditions – Failure	Deduction updates are not saved (e.g., invalid amounts).
Main Flow	1. User navigates to the Deductions management screen for a selected farmer. 2. System displays all current deductions, including auto-synced records from the mobile app (Advances, Distributed Items) and existing manual records. 3. User chooses to either edit a synced record OR add a new manual deduction. 4. If Editing: User modifies the amount or details (e.g., fixing a Collector's entry error) and inputs a mandatory revision note. 5. If Adding: User selects the deduction type (Loan, Special Deduction), enters the amount, and sets the installment plan. 6. User clicks "Save". 7. System validates the entry and updates the farmer's ledger in the master database.
Alternative Flow 1	In step 5: The limit is exceeded. If a manually added loan amount is suspiciously high, the system will flag it and require recheck before saving it.
Alternative Flow 2	At step 4: Attempting to edit an approved month. If the user tries to edit a deduction that belongs to a month where payments have already been approved (UC-15), the system blocks the edit and displays a "Record Locked" error.
Business Rules	BR-17: Scheduled and verified deductions are automatically pulled by UC-13 during monthly calculations. BR-18: Any modification made by Office Staff to a record originally synced from the mobile app (e.g., changing an Advance amount) must permanently log an audit trail showing the original amount, the new amount, the timestamp of the change, and the Staff ID.

3.4.12 UC - 12 – Set monthly tea rate

Use Case ID	UC-12
Use Case Name	Set monthly tea rate
Actor(s)	Owner / Office Staff
Preconditions	User has Owner/Admin privileges.
Postconditions – Success	The official price per KG is saved for the specified month.
Postconditions – Failure	Rate is not saved.
Main Flow	1. User selects "Set Tea Rate". 2. User selects the applicable Month/Year. 3. User inputs the final per-KG rate. 4. User confirms and saves. 5. System applies this rate globally for the selected month's calculation.
Alternative Flow 1	At step 4: Overwriting existing rate. If a rate already exists and payments are processed, the system prevents modifications unless a full recalculation is authorized.
Business Rules	BR-19: Only one active standard rate can exist per calendar month unless specific farmer-level tiering is enabled.

3.4.13 UC - 13 – Calculate Monthly Payment

Use Case ID	UC-13
Use Case Name	Calculate Monthly Payment (Includes UC-13a: Apply all deductions)
Actor(s)	Owner / Office Staff
Preconditions	The monthly tea rate (UC-12) has been set. All mobile devices have synced their end-of-month data.
Postconditions – Success	Net payable amounts are accurately calculated for all farmers, and the payment summary is ready for approval.
Postconditions – Failure	Calculation cannot proceed (e.g., missing tea rate or pending syncs alert).
Main Flow	1. Actor selects the target month and clicks "Run Calculations". 2. System verifies that the monthly tea rate is configured.

	3. System aggregates the total accepted leaf weight for each individual farmer. 4. System calculates Gross Income (Total Weight × Monthly Rate). 5. [UC-13a] System gathers all deductions for the month (Advances given, fertilizer costs, loan installments). 6. System subtracts total deductions from Gross Income to find the Net Payable. 7. System generates a preliminary Payment Summary Sheet for review. 8. System highlights any anomalies (e.g., negative balances).
Alternative Flow 1	At step 6: Negative Balance. If deductions exceed the gross income, the Net Payable becomes 0. The system automatically creates a carry-forward deduction record for the next month.
Alternative Flow 2	At step 2: Missing Rate. If the monthly rate is not set, system halts the process and prompts the actor to set the rate (UC-12) before continuing.
Business Rules	BR-20: Calculations must only include leaf records and advance records explicitly dated within the selected calendar month.

3.4.14 UC - 14 – View payment summary

Use Case ID	UC-14
Use Case Name	View payment summary
Actor(s)	Owner / Office Staff
Preconditions	UC-13 (Calculations) has been run for the month.
Postconditions – Success	System displays a comprehensive, filterable list of all farmer payments, gross incomes, deductions, and net pay.
Postconditions – Failure	Summary does not load.
Main Flow	1. User navigates to "Payment Summaries". 2. System retrieves calculation results for the current period. 3. System displays data in a tabulated grid. 4. User filters by Route or Collector to verify accuracy.
Business Rules	BR-21: Payment summary is read-only. Any required changes must be done via adjusting weights (UC-04) or deductions (UC-11) and re-running calculations (UC-13).

3.4.15 UC - 15 – Approve Payment Run

Use Case ID	UC-15
Use Case Name	Approve Payment Run
Actor(s)	Owner
Preconditions	UC-14 is reviewed and user has Owner privileges.
Postconditions – Success	Payment records locked as 'Approved'. Audit log updated. Bank transfer process (UC-12) triggered automatically.
Postconditions – Failure	Approval fails.
Main Flow	1. Owner clicks "Approve Payments" on the summary screen. 2. System prompts for secondary confirmation or admin PIN. 3. Owner enters confirmation. 4. System locks the month's records to prevent further edits.
Business Rules	BR-22: Once approved, no historical data (weights, rates, deductions) for that specific month can be altered.

3.4.16 UC - 16 – Process bank transfer

Use Case ID	UC-16
Use Case Name	Process bank transfer (Interacts with Bank API)
Actor(s)	Owner
Preconditions	Payment run is Approved (UC-15).
Postconditions – Success	API call to the bank is successful, and funds are scheduled for dispersal.
Postconditions – Failure	Bank API rejects the file/request.
Main Flow	1. Owner clicks "Process Bank Transfers". 2. System compiles net pay data and farmer bank details into the required Bank API format (or generates a bulk upload CSV/XML). 3. System transmits the payload via the secure Bank API. 4. System receives confirmation tokens from the bank. 5. System updates payment status to "Processing".
Alternative Flow 1	At step 4: Invalid accounts. Bank API returns errors for specific accounts (e.g., closed account). System flags these specific farmers for manual cash payment.
Business Rules	BR-23: All payloads must be encrypted and logged in an audit trail.

3.4.17 UC - 17 – Send SMS notification

Use Case ID	UC-17
Use Case Name	Send SMS notification (Interacts with SMS API)
Actor(s)	Owner / Office Staff
Preconditions	Payments are processed, or important alerts need sending.
Postconditions – Success	SMS messages are delivered via third-party API.
Postconditions – Failure	SMS gateway fails to send.
Main Flow	1. System detects a completed payment transfer OR User drafts a manual broadcast. 2. System parses farmer phone numbers. 3. System sends payloads to the SMS Gateway API. 4. System logs delivery statuses.
Alternative Flow 1	At step 3: API Timeout. System retries up to 3 times before logging a failure.
Business Rules	BR-24: Automated payment SMS must include Total Weight, Rate, Total Deductions, and Net Paid. BR-25: SMS notifications shall be sent only after a bank transfer is confirmed successful.

3.4.18 UC - 18 – Generate reports

Use Case ID	UC-18
Use Case Name	Generate reports
Actor(s)	Owner / Office Staff
Preconditions	System contains collection and payment data.
Postconditions – Success	Report (PDF/Excel) is generated and downloaded.
Postconditions – Failure	Report fails to compile.
Main Flow	1. User navigates to "Reports". 2. User selects report type (e.g., Monthly Collection by Route, Deduction Totals). 3. User sets date range parameters. 4. System queries database and compiles the visual/tabular report. 5. User exports or prints.
Business Rules	BR-26: All generated reports must contain a timestamp of generation for auditing.

3.4.19 UC - 19 – Manage user accounts

Use Case ID	UC-19
Use Case Name	Manage user accounts
Actor(s)	Owner
Preconditions	User is logged in as Owner.
Postconditions – Success	Collector or Staff account is created/modified/disabled.
Postconditions – Failure	Account is not modified.
Main Flow	1. Owner views user list. 2. Owner selects "Add User". 3. Owner assigns Role (Collector vs Staff), Username, and default password. 4. System saves the new credentials to the authentication server.
Alternative Flow 1	At step 4: Weak password. System rejects passwords under 8 characters.
Business Rules	BR-27: Usernames must be unique across the entire system.

3.4.20 UC - 20 – Manage collection routes

Use Case ID	UC-20
Use Case Name	Manage collection routes
Actor(s)	Owner / Office Staff
Preconditions	User is logged in.
Postconditions – Success	Routes are created or modified, and farmers are reassigned.
Postconditions – Failure	Route is not saved.
Main Flow	1. User selects "Route Management" from the dashboard. 2. User clicks "Create New Route" or selects an existing route to edit. 3. User enters or updates the Route Name (e.g., "North Valley"). 4. User assigns an active Tea Collector to be responsible for this route. 5. System saves the route details to the master database.
Alternative Flow 1	At step 4: Collector reassignment. If the selected Collector is currently assigned to another active route, the system displays a warning prompt. The user can confirm to reassign the Collector to the new route or cancel.
Business Rules	BR-28: A farmer can only belong to one active collection route at a time. BR-29: A route must be successfully created here before it can populate the dropdown menu used during "Register New Farmer" (UC-09). BR-30: If a route is marked as "Deactivated", the system should mark all farmers assigned to that route so that they can be reassigned.

3.5 Activity Diagrams

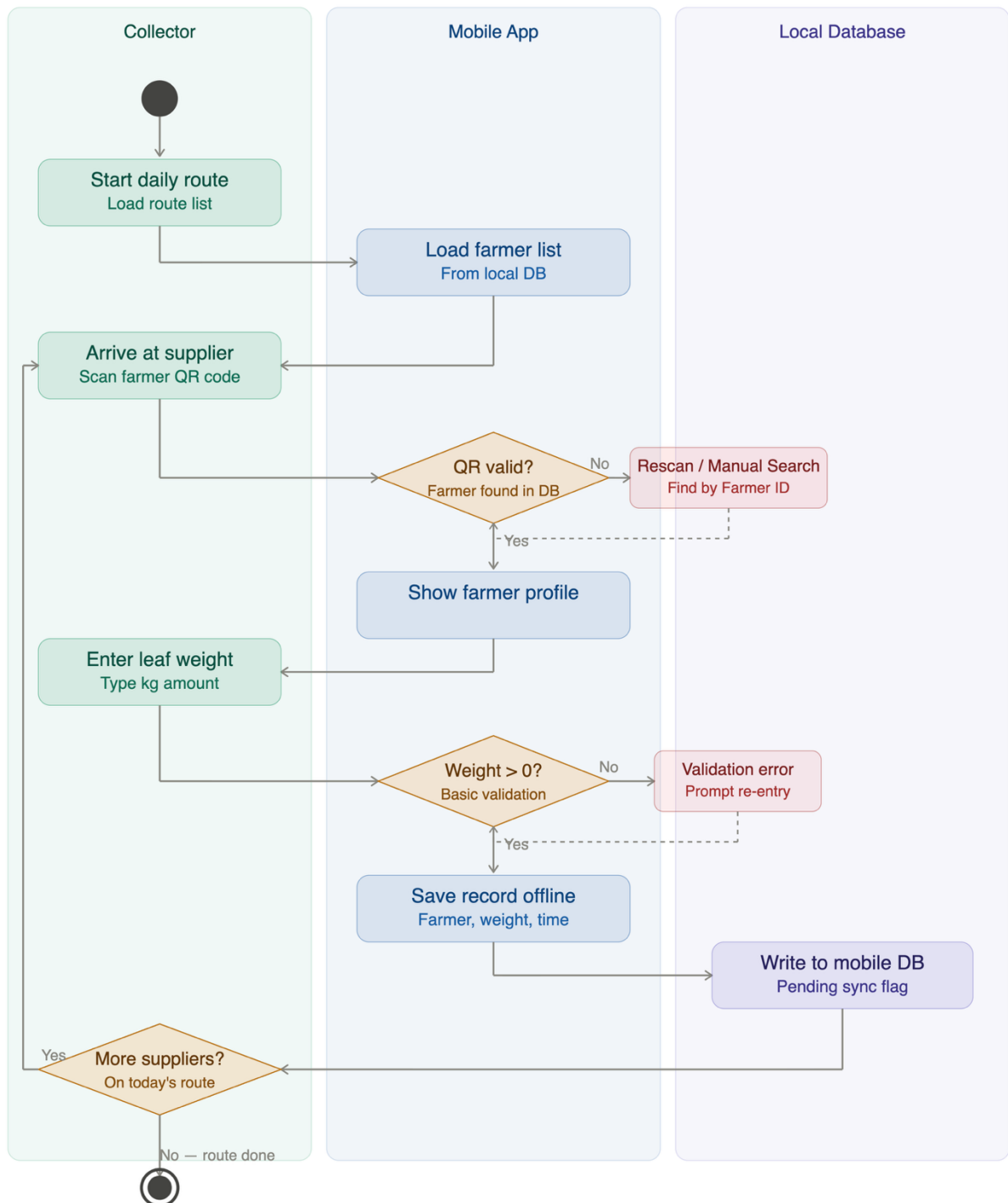
Activity diagrams are provided for the five most critical use cases.

3.5.1 Daily Tea Leaf Collection

This diagram maps out the core offline-first field operations performed by the Tea Collector. It illustrates the sequence of starting a daily route, scanning a farmer's QR code, validating and entering the leaf weight, and securely saving the transaction to the mobile device's local database without requiring an active internet connection.

Use Cases Covered:

- **UC-02:** View assigned route
- **UC-03:** Scan farmer QR code
- **UC-04:** Record leaf weight (Includes **UC-04a:** Save record offline)

Activity Diagram 1 – Daily tea leaf collection

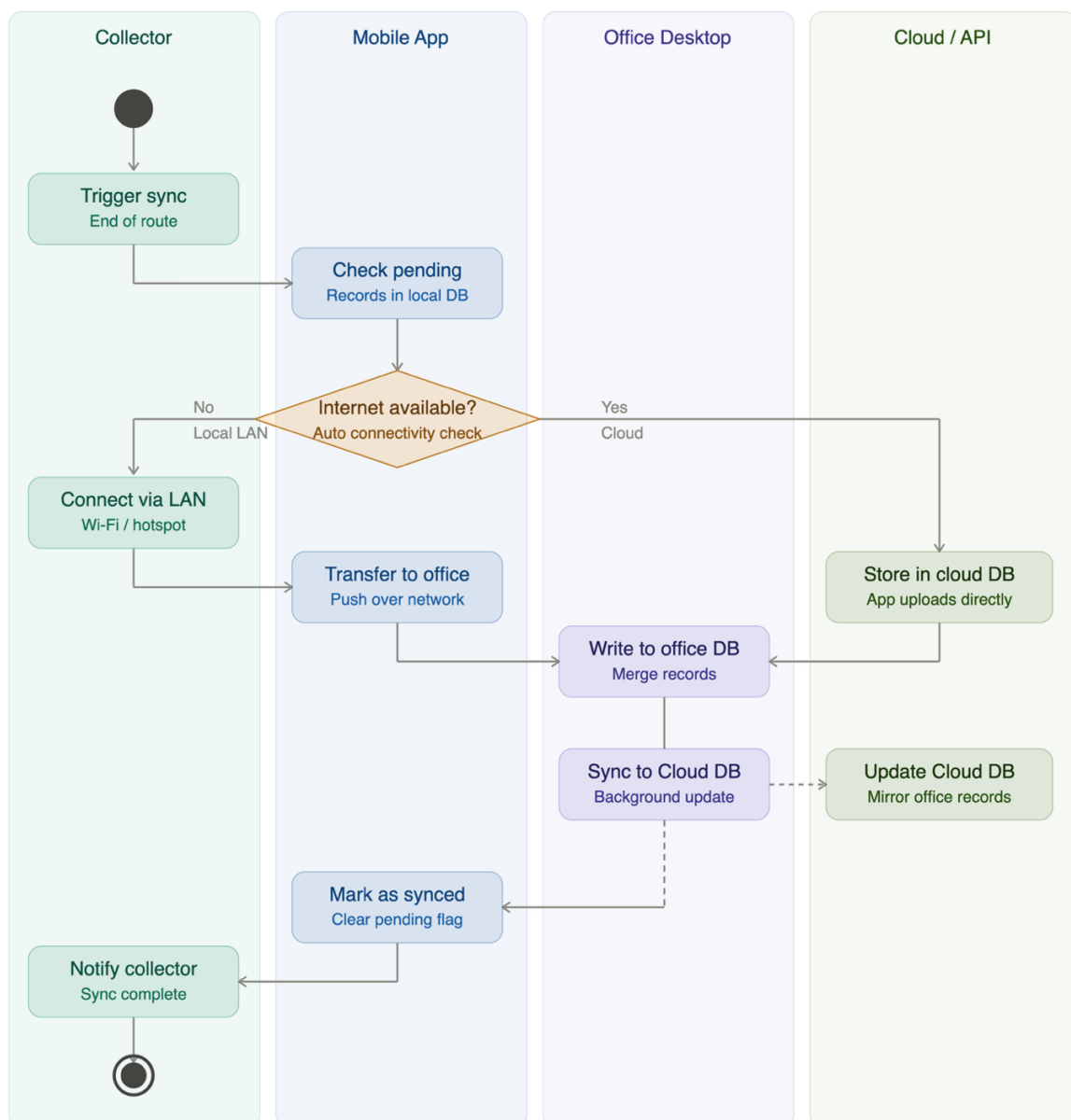
3.5.2 Data Synchronization (Local and Cloud)

This diagram details the distributed data architecture of the system, showcasing the decision-making process for syncing mobile data. It demonstrates how the system detects connectivity and dynamically routes unsynced local data either directly to the Cloud database (via mobile internet) or through the local office network (LAN) to ensure data integrity.

Use Cases Covered:

- **UC-07:** Sync collection data
- **UC-07a:** Local LAN sync (Extended)
- **UC-07b:** Cloud sync (Extended)

Activity Diagram 2 – Data synchronisation (local and cloud)



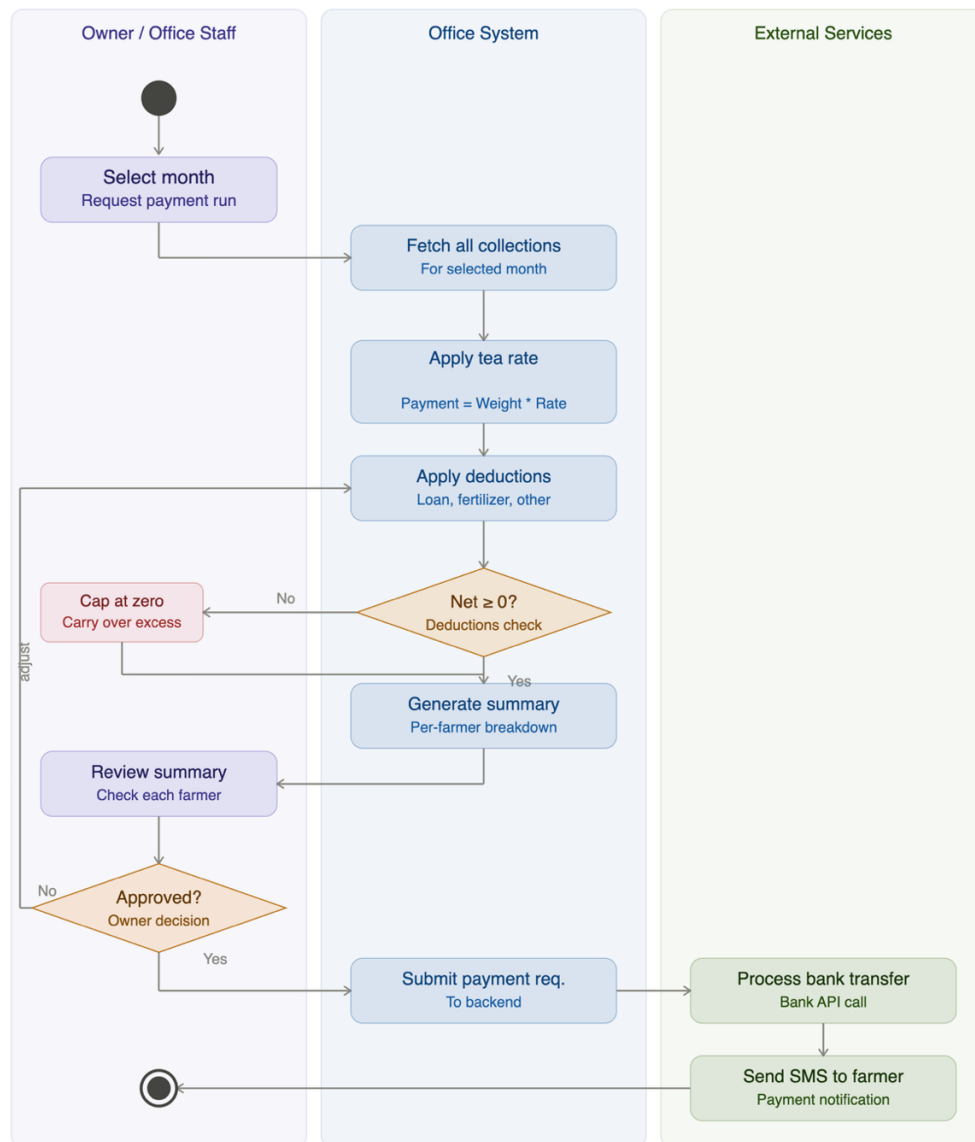
3.5.3 Monthly Payment Calculation and Processing

This diagram visualizes the heavy financial "engine" of the desktop application. It tracks the end-of-month workflow from aggregating leaf weights and applying the standard tea rate, to deducting advances/loans, handling negative net balances, securing owner approval, and finally communicating with external Bank and SMS APIs to execute the payments.

Use Cases Covered:

- **UC-13:** Calculate monthly payment (Includes **UC-13a:** Apply all deductions)
- **UC-14:** View payment summary
- **UC-15:** Approve payment run
- **UC-16:** Process bank transfer (API)
- **UC-17:** Send SMS notification (API)

Activity Diagram 3 – Monthly payment calculation and processing



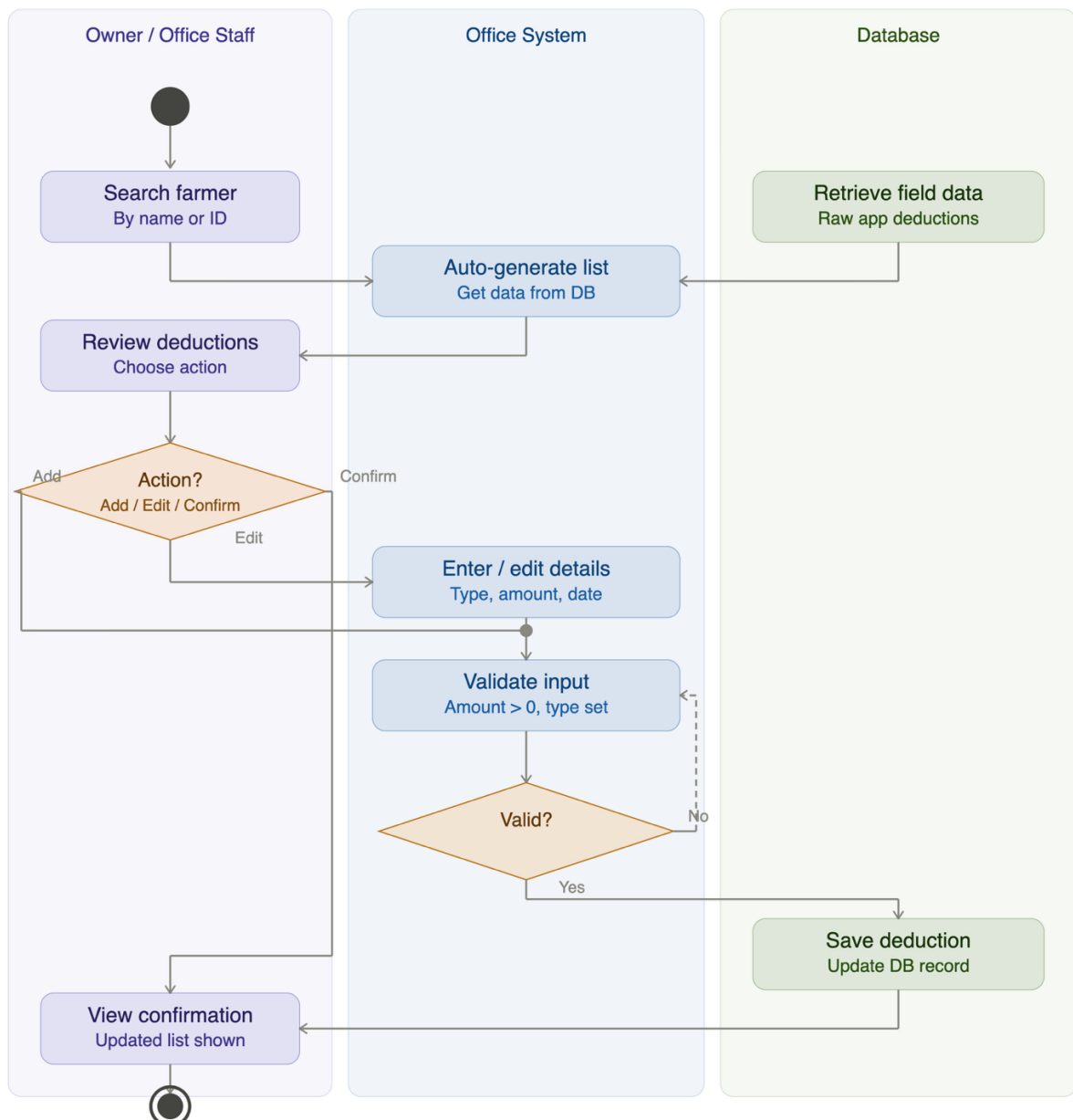
3.5.4 Deduction Management (Loans and Fertilizer)

This diagram explains the administrative workflow for managing farmer liabilities. It shows how the office system aggregates raw deduction data from the field (like distributed fertilizer or cash advances) and allows the office staff to review, manually add, edit, or confirm these deductions against a farmer's profile before the final monthly calculations are run.

Use Cases Covered:

- **UC-11:** Manage deductions
- **UC-11a / UC-11b / UC-11c:** Add loan, fertilizer, and other deductions (Extended)

Activity Diagram 4 – Deduction management (loans and fertilizer)



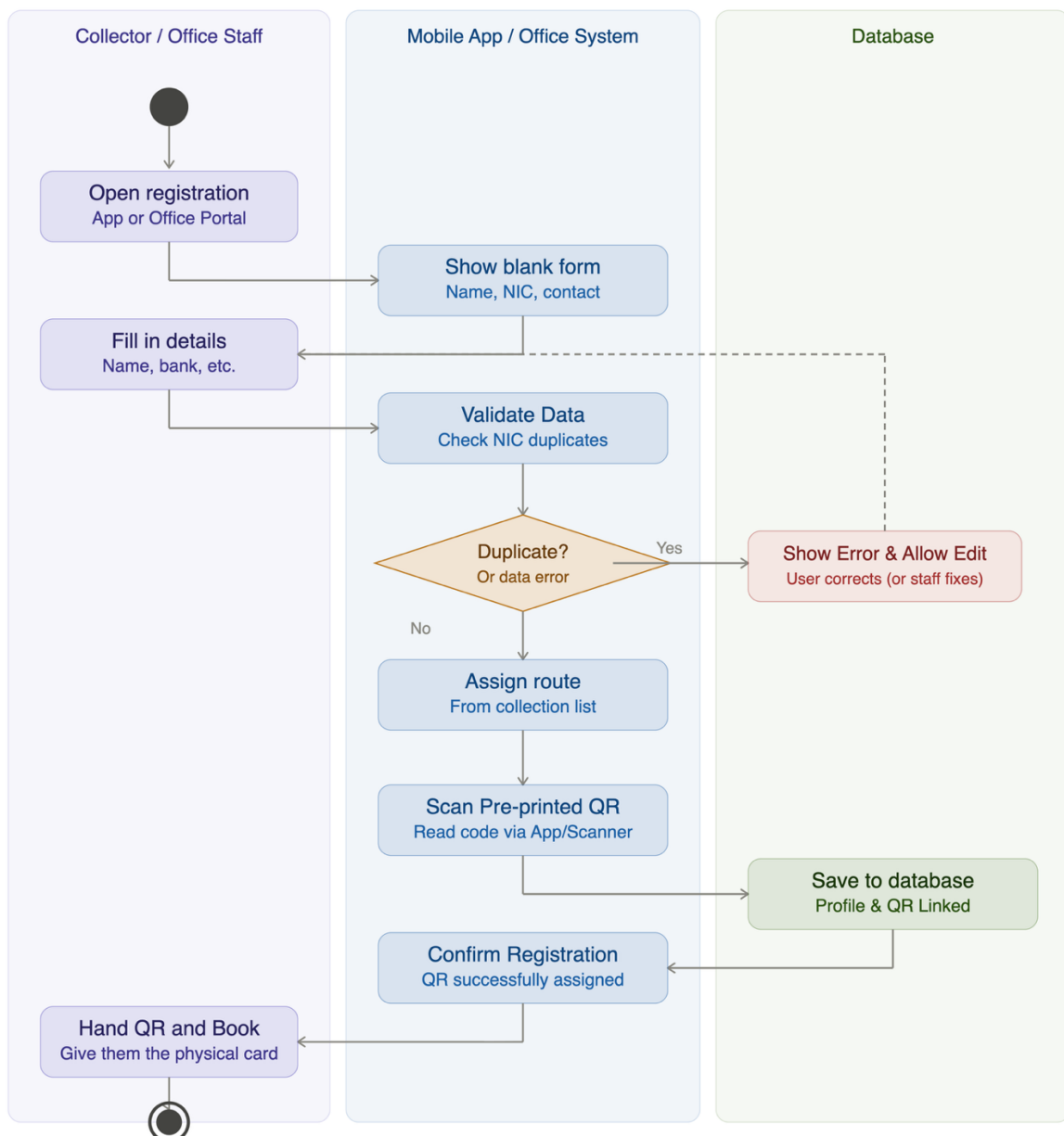
3.5.5 Farmer Registration and Profile Management

This diagram illustrates the onboarding process for new suppliers, highlighting a flow that can be executed by either the Collector (via mobile) or Office Staff (via desktop). It outlines the entry of biographical data, validation against duplicate records (like NIC checks), route assignment, and the physical linking of a pre-printed QR code card to the newly created digital profile.

Use Cases Covered:

- **UC-09:** Register new farmer
- **UC-09a:** Generate or add QR code (Included)
- **UC-21:** Receive QR code card (Physical external actor action)

Activity Diagram 5 – Farmer registration and profile management



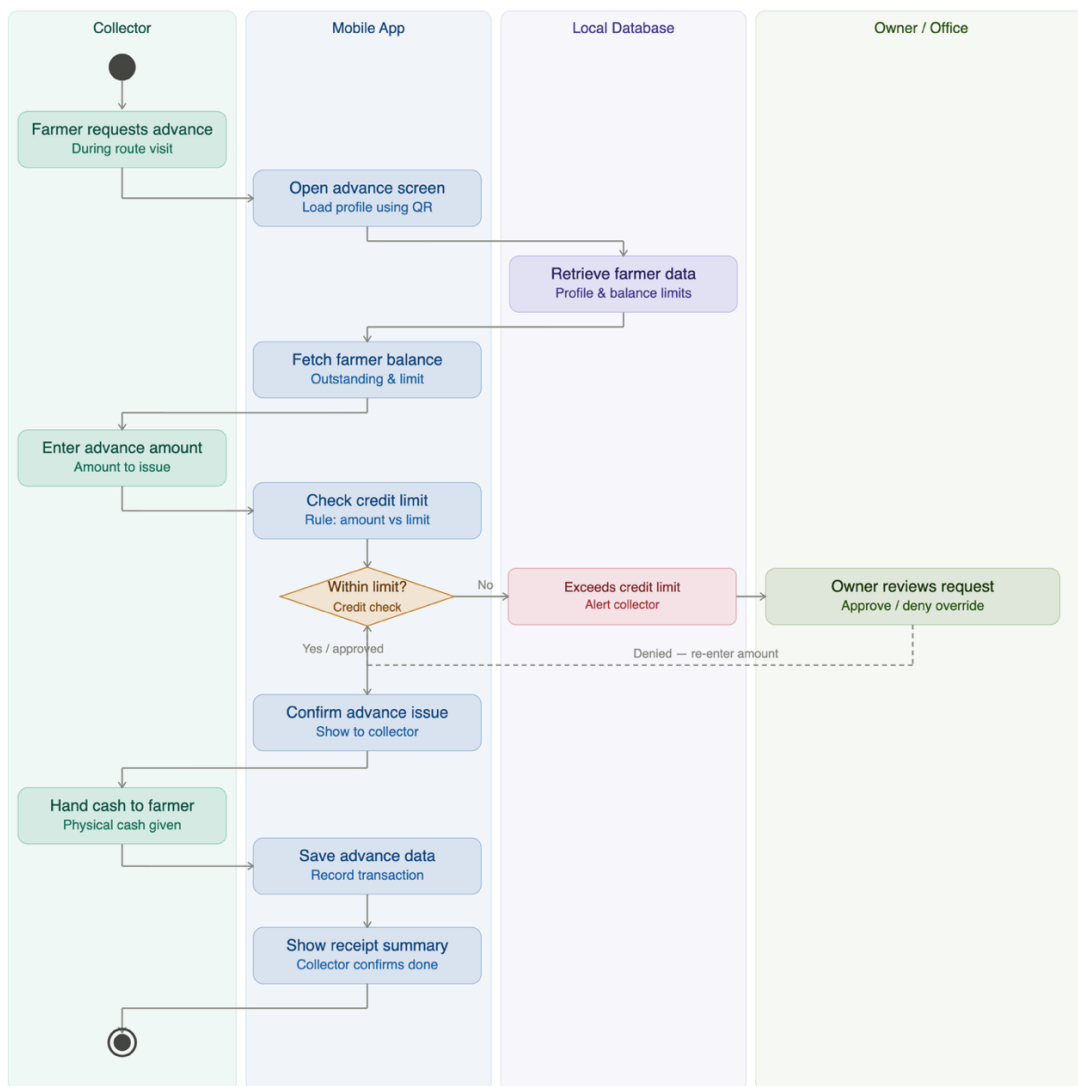
3.5.6 Advance Payment

This diagram illustrates the workflow for handing out assets—whether issuing cash advances or distributing physical items like fertilizer—to farmers in the field. It highlights the mobile app's ability to query the local database to verify a farmer's accumulated balance and credit limits before approving the transaction. It also maps out the exception-handling process for when a request exceeds the limit (requiring an override review by the Owner) and concludes with the physical handover and secure offline saving of the transaction.

Use Cases Covered:

- **UC-05:** Item distribution and record keeping
- **UC-05a:** Save record offline (Included)
- **UC-06:** Issue advance payment and record keeping

Activity Diagram 6 — Advance payment

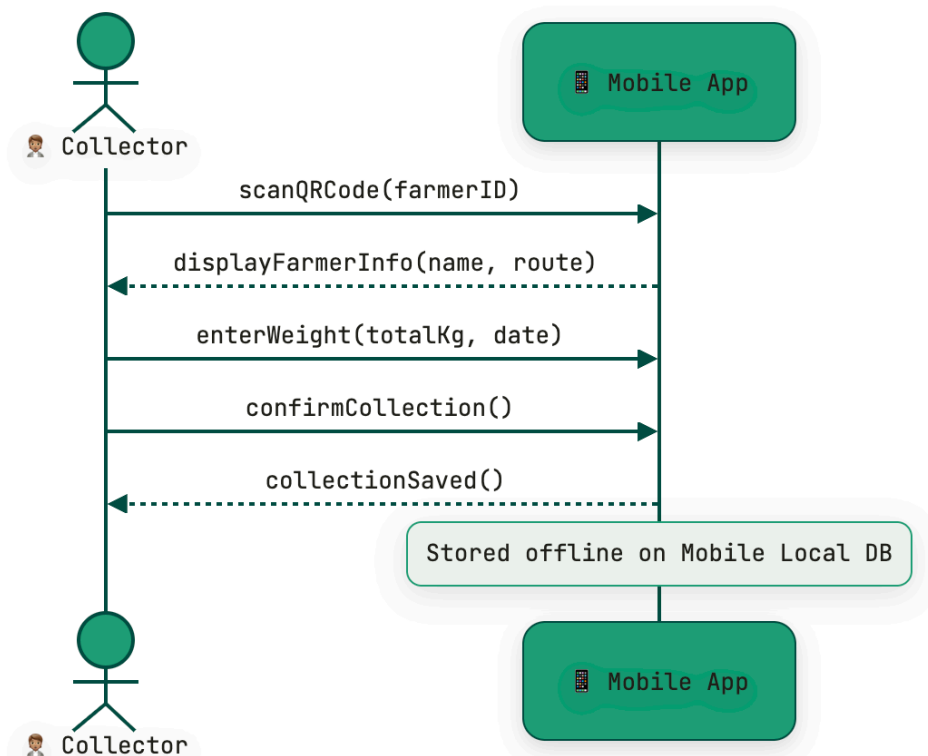


3.6 System Sequence Diagrams (High-Level)

System Sequence Diagrams (SSDs) show the interaction between actors and the TeaRoutePay system boundary for the key use cases.

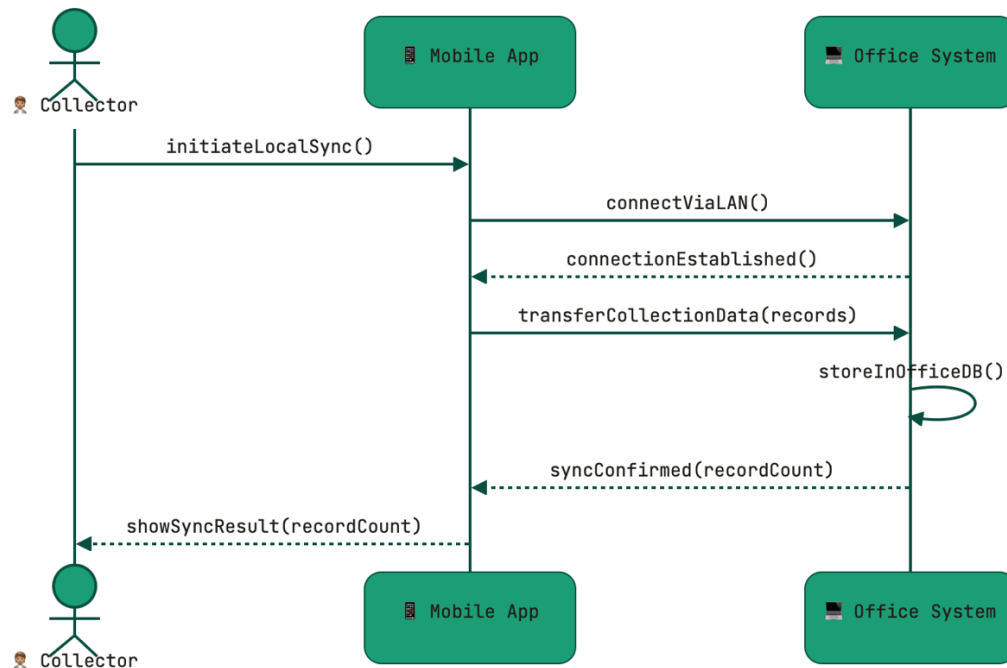
3.6.1 SSD - 01 — Record Tea Collection

The Collector uses the mobile app to record daily tea leaf weights at each farmer stop along the route. The farmer is identified via QR code scan, and the weight entry is saved offline on the device immediately — no internet connection is required in the field.



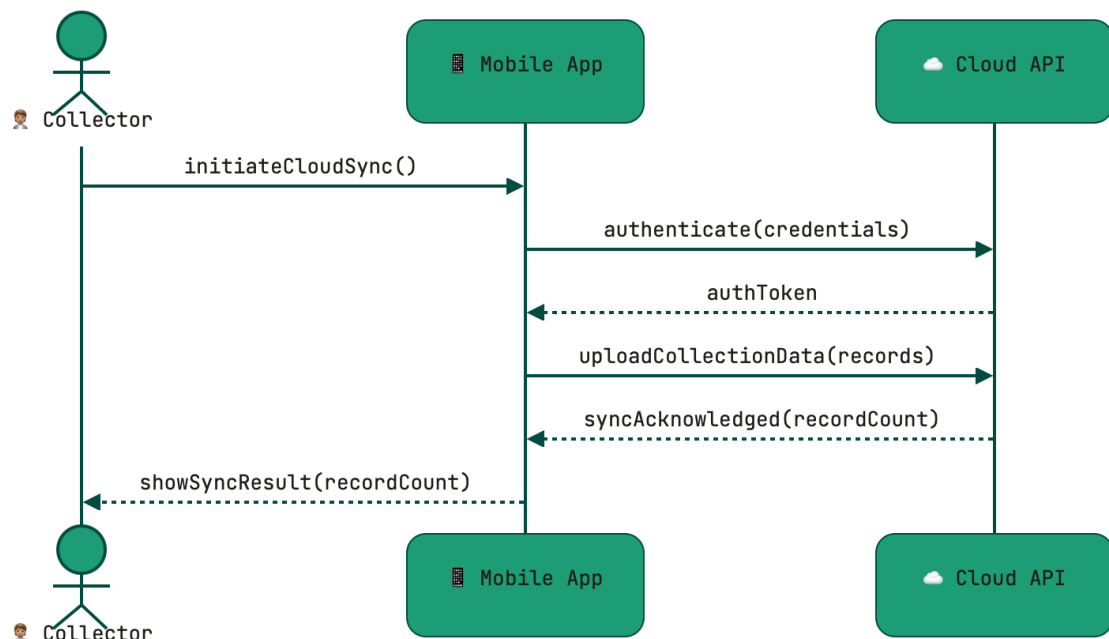
3.6.2 SSD - 02 — Sync Data (Local Mode)

After completing the route, the Collector selects local sync mode. The mobile app connects to the main office computer via the local area network and transfers all buffered collection records in bulk, making them available to office staff immediately.



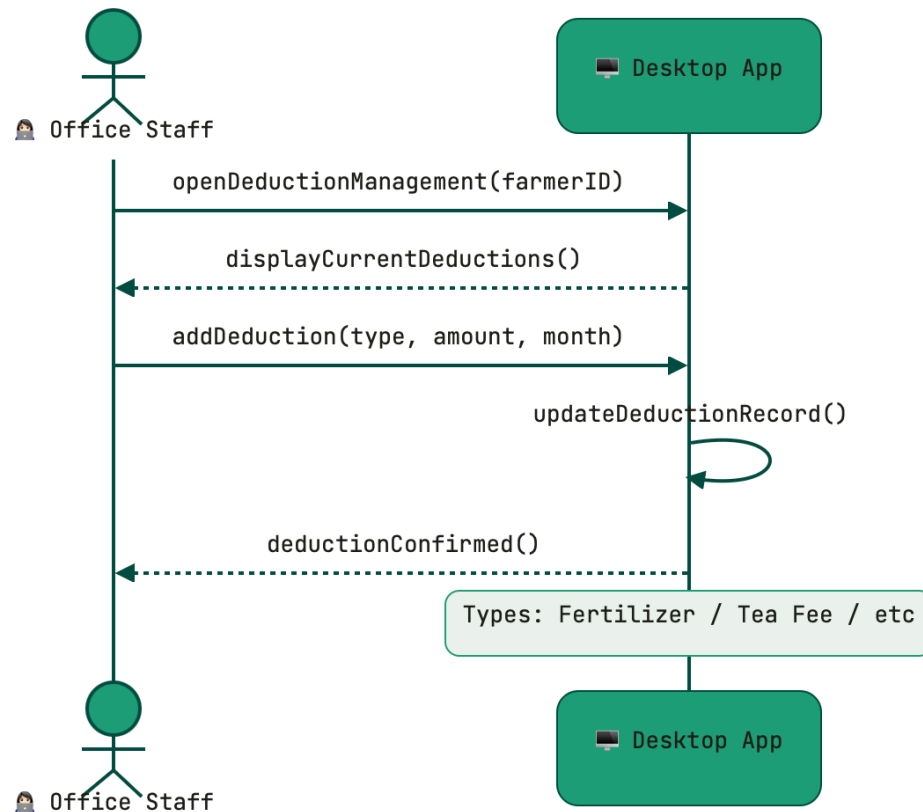
3.6.3 SSD - 03 — Sync Data (Cloud Mode)

When cloud mode is selected (or the collector is remote), the mobile app authenticates with the cloud backend and uploads all pending collection records. The cloud database is updated, making the data accessible to the office desktop app from any location.



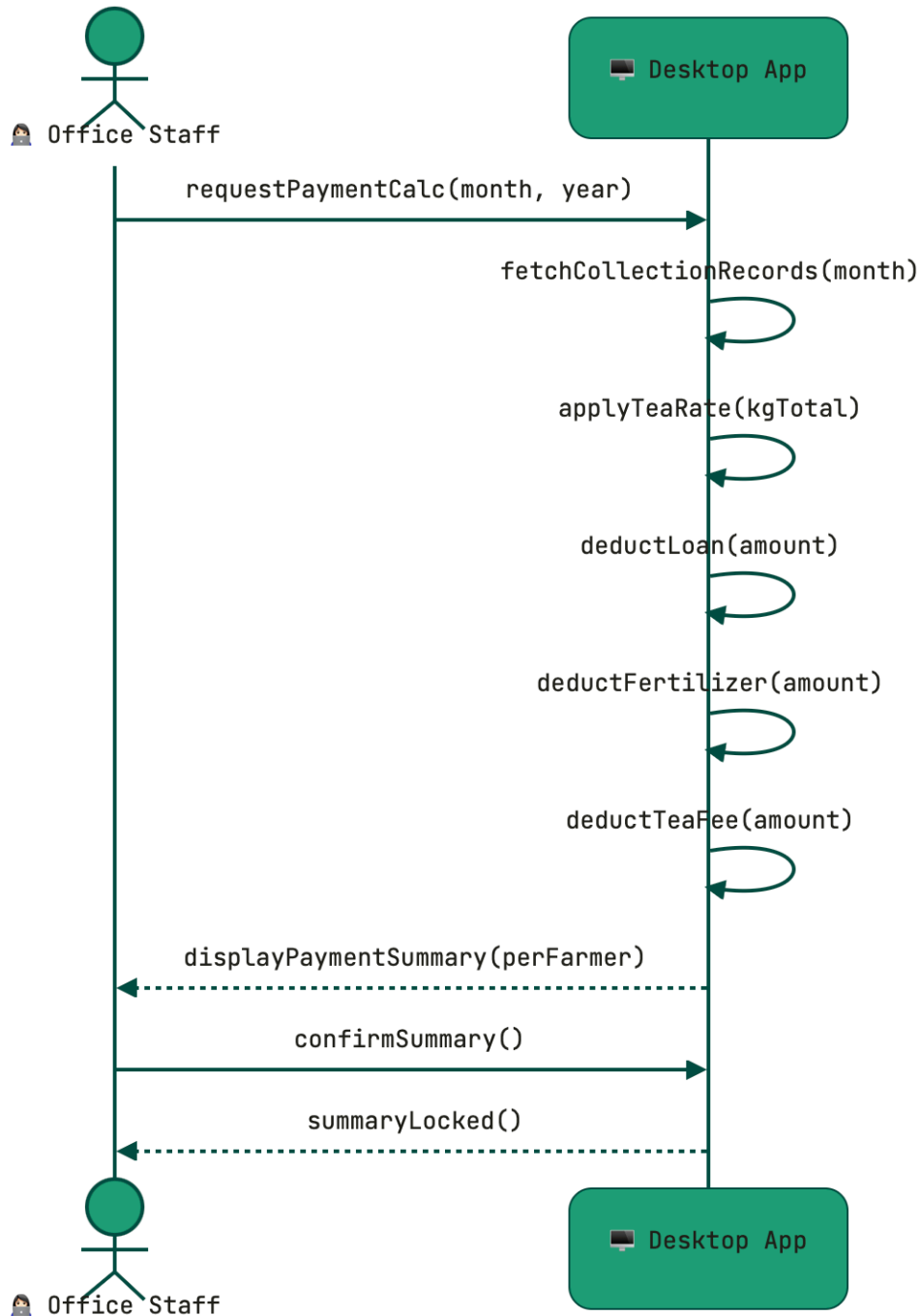
3.6.4 SSD - 04 — Manage Deductions

Office staff records all applicable deductions against a farmer's account. These are stored and automatically applied when monthly payment calculations are run.



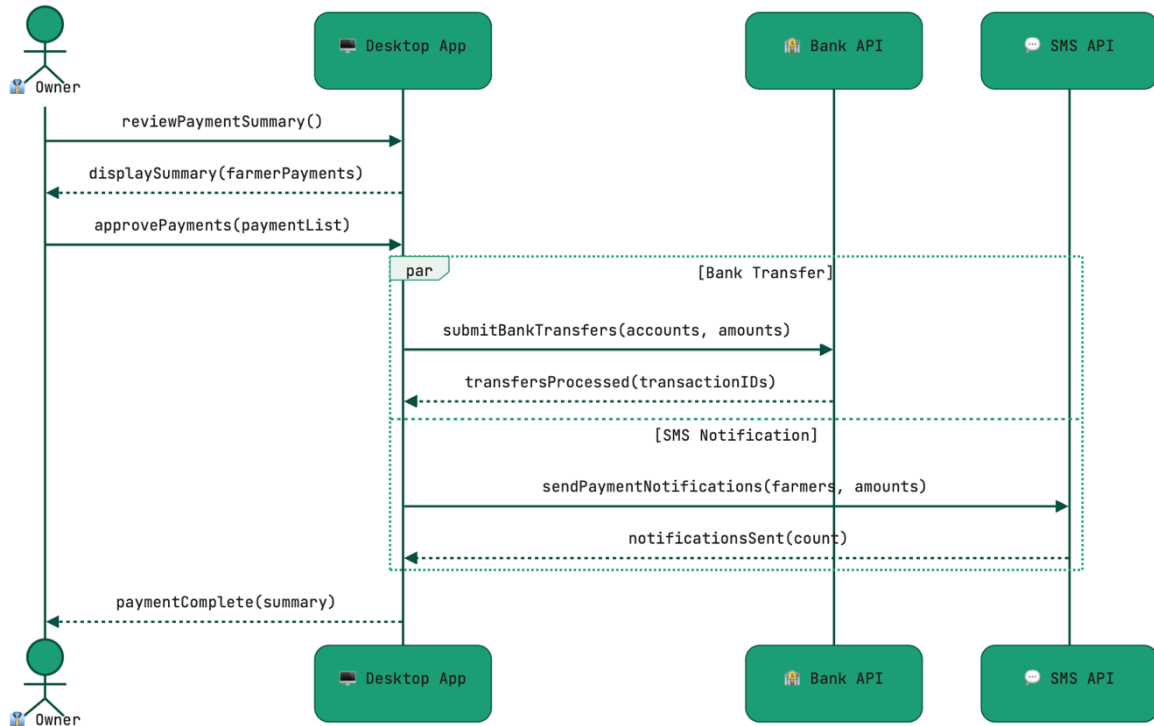
3.6.5 SSD - 05 — Calculate Monthly Payment

At month end, office staff requests the monthly payment calculation. The system aggregates all collection weights for that month, applies the current tea rate, then subtracts each farmer's applicable deductions — producing a per-farmer net payable summary ready for Owner review.



3.6.6 SSD - 06 — Approve & Process Payment

The Owner reviews and approves the locked payment summary prepared by office staff. On approval, the system submits bulk bank transfer instructions to the bank API for all eligible farmers and triggers SMS notifications confirming each farmer's net payment amount.



4. Functional Requirements

Priority scale:

P1 – Critical (system cannot function without this).
 P2 – High (important for full operation).
 P3 – Medium (valuable enhancement).
 P4 – Low (deferred to later release).

4.1 User Authentication and Authorization

ID	FR-01
Title	User Login
Description	The system shall authenticate users with a username and password before granting access to any function.
Priority	P1 – Critical
Source	Client Interview #1; UC-01, UC-08
Verification	Test Case TC-FR-01 (functional test)

ID	FR-02
Title	Role-Based Access Control
Description	The system shall enforce role-based access control with four roles: Owner, Office Staff, Collector, and System Administrator. Each role shall have a distinct, non-overlapping permission set.
Priority	P1 – Critical
Source	Client Interview #1; UC-01, UC-08
Verification	Test Case TC-FR-02 (security test)

ID	FR-03
Title	Account Lockout
Description	The system shall lock a user account after five consecutive failed login attempts. A System Administrator must manually unlock the account.
Priority	P2 – High
Source	Security Review; NFR-30
Verification	Test Case TC-FR-03 (security test)

ID	FR-04
Title	Session Timeout
Description	The system shall automatically expire sessions after a configurable idle timeout period. The default idle timeout shall be 30 minutes.
Priority	P2 – High
Source	Security Review
Verification	Test Case TC-FR-04 (functional test)

ID	FR-05
Title	User Account Management
Description	Owner shall be able to create, update, deactivate, and reset passwords for user accounts.
Priority	P2 – High
Source	Client Interview #1; UC-19
Verification	Test Case TC-FR-05 (functional test)

ID	FR-06
Title	Audit Logging of Authentication Events
Description	All authentication events (login success, login failure, logout, account lock) shall be written to an immutable security audit log with user ID and timestamp.
Priority	P2 – High
Source	Security Review
Verification	Inspection TC-FR-06 (log review)

4.2 Farmer Management

ID	FR-07
Title	Farmer Registration
Description	The system shall allow Office Staff and collector to create a farmer profile with mandatory fields: full name, NIC number, address, primary phone number, route assignment, and bank account number.
Priority	P1 – Critical
Source	Client Interview #1; UC-09
Verification	Test Case TC-FR-07 (functional test)

ID	FR-08
Title	Unique NIC Validation
Description	The system shall validate that the NIC number is unique across all farmer records (including deactivated records) and reject duplicate entries.
Priority	P1 – Critical
Source	Document Analysis
Verification	Test Case TC-FR-08 (validation test)

ID	FR-09
Title	QR Code Generation
Description	The system shall automatically generate(add) and display a unique QR code for each registered farmer upon profile creation. The QR code shall encode the farmer UUID only.
Priority	P1 – Critical
Source	On-Site Observation; UC-03, UC-21
Verification	Test Case TC-FR-09 (functional test)

ID	FR-10
Title	QR Card Printing
Description	The system shall allow Office Staff to print individual farmer QR code cards from the farmer profile screen.
Priority	P2 – High
Source	On-Site Observation
Verification	Test Case TC-FR-10 (functional test)

ID	FR-11
Title	Farmer Profile Update
Description	The system shall allow Office Staff to update any editable farmer profile field. All changes shall be logged in the audit trail with the performing user's ID and timestamp.
Priority	P2 – High
Source	Client Interview #2; UC-10
Verification	Test Case TC-FR-11 (functional test)

ID	FR-12
Title	Farmer Deactivation
Description	The system shall allow soft-deletion (deactivation) of farmer profiles. Deactivated farmers shall not appear in active collection lists. A farmer with collection or payment history cannot be hard-deleted.
Priority	P2 – High
Source	Client Interview #2; UC-10
Verification	Test Case TC-FR-12 (functional test)

ID	FR-13
Title	Farmer Search and Filter
Description	The system shall provide search and filter of farmers by name, NIC, route assignment, or active status.
Priority	P3 – Medium
Source	Client Interview #2
Verification	Test Case TC-FR-13 (functional test)

4.3 Route Management

ID	FR-14
Title	Route Creation
Description	The system shall allow the Owner to create collection routes with a route name, route code, and an assigned Collector.
Priority	P1 – Critical
Source	On-Site Observation; UC-20
Verification	Test Case TC-FR-14 (functional test)

ID	FR-15
Title	Farmer-to-Route Assignment
Description	The system shall allow multiple farmers to be assigned to a single route. Farmers may be re-assigned between routes; the full assignment history shall be retained.
Priority	P1 – Critical
Source	On-Site Observation
Verification	Test Case TC-FR-15 (functional test)

ID	FR-16
Title	Route Schedule
Description	The system shall support daily route scheduling – specifying which route runs on which days of the week.
Priority	P3 – Medium
Source	Questionnaire
Verification	Test Case TC-FR-16 (functional test)

4.4 Field Collection – Mobile Application

ID	FR-17
Title	Route Selection at Session Start
Description	The mobile app shall allow a Collector to select their assigned route before beginning a collection session.
Priority	P1 – Critical
Source	On-Site Observation; UC-02
Verification	Test Case TC-FR-17 (functional test)

ID	FR-18
Title	QR Code Scanning
Description	The mobile app shall scan a farmer QR code using the device camera to identify the farmer for a collection entry.
Priority	P1 – Critical
Source	On-Site Observation; UC-03
Verification	Test Case TC-FR-18 (functional test)

ID	FR-19
Title	Manual Farmer Fallback Selection
Description	The mobile app shall allow manual farmer selection by name or ID as a fallback when QR scanning is not possible.
Priority	P1 – Critical
Source	Questionnaire
Verification	Test Case TC-FR-19 (functional test)

ID	FR-20
Title	Offline Weight Entry and Local Storage
Description	The mobile app shall allow the Collector to enter tea leaf weight and store the record locally, regardless of network availability.
Priority	P1 – Critical
Source	On-Site Observation; UC-04
Verification	Test Case TC-FR-20 (offline test)

ID	FR-21
Title	Weight Entry Correction (Pre-Sync)
Description	The mobile app shall allow editing of a collection weight entry before synchronisation. The original value and the correcting user shall be retained in the record.
Priority	P2 – High
Source	Client Interview #2;
Verification	Test Case TC-FR-21 (functional test)

ID	FR-22
Title	Duplicate Entry Prevention
Description	The mobile app shall detect a potential duplicate entry for the same farmer on the same date and session, and require explicit Collector confirmation before saving.
Priority	P2 – High
Source	Document Analysis; BR-06
Verification	Test Case TC-FR-22 (validation test)

ID	FR-23
Title	Session Summary View
Description	The mobile app shall display a running summary of total kilograms collected per route for the current session, updated after each entry.
Priority	P3 – Medium
Source	Questionnaire
Verification	Test Case TC-FR-23 (functional test)

ID	FR-24
Title	Connectivity Status Indicator
Description	The mobile app shall display a visual indicator showing the current connectivity status: Offline, LAN Connected, or Internet Connected.
Priority	P3 – Medium
Source	Questionnaire
Verification	Test Case TC-FR-24 (functional test)

4.5 Data Synchronization

ID	FR-25
Title	Dual Sync Mode Support
Description	The system shall support two synchronisation modes: Local LAN Sync and Cloud Sync. The active mode shall be configurable by the Owner without requiring a code change.
Priority	P1 – Critical
Source	Client Interview #1; UC-07
Verification	Test Case TC-FR-25 (configuration test)

ID	FR-26
Title	Local LAN Synchronisation
Description	In Local LAN Sync mode, the mobile app shall transfer all pending collection records to the Office Local Database when connected to the office LAN.
Priority	P1 – Critical
Source	Client Interview #1; UC-07a
Verification	Test Case TC-FR-26 (integration test)

ID	FR-27
Title	Cloud Synchronisation
Description	In Cloud Sync mode, the mobile app shall upload all pending collection records to the Backend API Server over the Internet.
Priority	P1 – Critical
Source	Client Interview #1; UC-07b
Verification	Test Case TC-FR-27 (integration test)

ID	FR-28
Title	Sync Conflict Resolution
Description	The system shall detect data conflicts (same farmer, same date, different weight from two devices) and resolve using a last-write-wins strategy based on device timestamp. All conflicts shall be logged for owner review.
Priority	P2 – High
Source	Document Analysis; BR-06, BR-10
Verification	Inspection TC-FR-28 (conflict log review)

ID	FR-29
Title	Sync Retry with Back-Off
Description	The system shall retry failed synchronisation attempts up to three times using exponential back-off before notifying the user of failure.
Priority	P2 – High
Source	Technical Constraint; BR-11
Verification	Test Case TC-FR-29 (resilience test)

4.6 Tea Rate Management

ID	FR-30
Title	Monthly Tea Rate Setting
Description	The system shall allow the Owner or Office Staff to set the monthly tea purchase rate (LKR per kg) for each calendar month.
Priority	P1 – Critical
Source	Client Interview #1; UC-12
Verification	Test Case TC-FR-30 (functional test)

ID	FR-31
Title	Tea Rate History
Description	The system shall maintain a history of all monthly tea rates. Past rates for months with completed payment calculations shall not be editable.
Priority	P2 – High
Source	Document Analysis
Verification	Inspection TC-FR-31 (data integrity test)

4.7 Deduction Management

ID	FR-32
Title	Loan Deduction Recording
Description	The system shall allow Office Staff to record a loan issued to a farmer with: total loan amount, monthly installment amount, start month, and description.
Priority	P2 – High
Source	Client Interview #2; UC-11
Verification	Test Case TC-FR-32 (functional test)

ID	FR-33
Title	Automatic Loan Installment Deduction
Description	The system shall automatically apply the monthly loan installment deduction during payment calculation until the loan balance reaches zero.
Priority	P2 – High
Source	Client Interview #2 / UC-11a
Verification	Test Case TC-FR-33 (calculation test)

ID	FR-34
Title	Fertilizer Deduction Recording
Description	The system shall allow Office Staff to record fertilizer or input material supplies issued to a farmer as a one-time or recurring deduction.
Priority	P2 – High
Source	Client Interview #2 / UC-11b, UC05
Verification	Test Case TC-FR-34 (functional test)

ID	FR-35
Title	Custom Deduction Types
Description	The System Administrator shall be able to define additional custom deduction types beyond the built-in Loan and Fertilizer types.
Priority	P3 – Medium
Source	Client Interview #2; UC-11
Verification	Test Case TC-FR-35 (functional test)

ID	FR-36
Title	Negative Payment Alert
Description	The system shall alert Office Staff when a farmer's net payment would be negative after applying deductions, and prevent the payment run from proceeding to the approval stage until the issue is resolved.
Priority	P1 – Critical
Source	Document Analysis; UC-13 Alternative Flow 1
Verification	Test Case TC-FR-36 (validation test)

4.8 Monthly Payment Calculation

ID	FR-37
Title	Gross Payment Calculation
Description	The system shall calculate the monthly gross payment per farmer as: Gross Payment = Total kg Collected (for the month) × Monthly Tea Rate, using fixed-point arithmetic.
Priority	P1 – Critical
Source	Client Interview #1/ UC-13
Verification	Test Case TC-FR-37 (calculation test)

ID	FR-38
Title	Net Payment Calculation
Description	The system shall calculate the monthly net payment per farmer as: Net Payment = Gross Payment – Sum of all applicable deductions for the month.
Priority	P1 – Critical
Source	Client Interview #1/ UC-13a
Verification	Test Case TC-FR-38 (calculation test)

ID	FR-39
Title	Payment Preview
Description	The system shall display a payment preview table showing for each farmer: total kg, gross payment (LKR), itemised deductions (LKR), and net payment (LKR), before the run is sent for approval.
Priority	P1 – Critical
Source	Client Interview #1/ UC-14
Verification	Test Case TC-FR-39 (functional test)

4.9 Payment Approval Workflow

ID	FR-40
Title	Owner Payment Approval
Description	The Owner shall be able to review the monthly payment preview and approve or reject the payment run. A rejected run must include a rejection reason.
Priority	P1 – Critical
Source	Client Interview #1; UC-15
Verification	Test Case TC-FR-40 (functional test)

ID	FR-41
Title	Payment Record Immutability on Approval
Description	Once a payment run is approved by the Owner, all associated payment records shall be locked and immutable. No further edits shall be permitted by any user.
Priority	P1 – Critical
Source	Document Analysis; BR-22
Verification	Inspection TC-FR-41 (security test)

ID	FR-42
Title	Approval Audit Record
Description	The system shall record the approving Owner's identity, approval timestamp, month, total approved amount, and any remarks against each approved payment run in the audit log.
Priority	P2 – High
Source	Document Analysis; BR-22
Verification	Inspection TC-FR-42 (log review)

4.10 Bank Transfer

ID	FR-43
Title	Bank Transfer Submission
Description	The system shall submit approved payments to the Bank Transfer API within 5 minutes of Owner approval.
Priority	P1 – Critical
Source	Client Interview #1; UC-16, BR-23
Verification	Test Case TC-FR-43 (integration test)

ID	FR-44
Title	Transfer Status Retrieval
Description	The system shall retrieve and display the bank transfer status response (Success / Pending / Failed) from the Bank Transfer API for each payment.
Priority	P1 – Critical
Source	Client Interview #1;
Verification	Test Case TC-FR-44 (integration test)

ID	FR-45
Title	Transfer Audit Log
Description	All bank transfer API requests, responses, HTTP status codes, and timestamps shall be stored in the audit log.
Priority	P1 – Critical
Source	Security Review; BR-23
Verification	Inspection TC-FR-45 (log review)

ID	FR-46
Title	Manual Transfer Re-submission
Description	The system shall allow the Owner or Office Staff to manually re-submit a failed bank transfer after the failure cause has been resolved.
Priority	P2 – High
Source	Client Interview #2
Verification	Test Case TC-FR-46 (functional test)

4.11 SMS Notification

ID	FR-47
Title	Farmer SMS on Successful Payment
Description	The system shall send an SMS to each farmer upon confirmation of a successful bank transfer, stating the net payment amount in LKR and the payment month.
Priority	P2 – High
Source	Client Interview #1; UC-17
Verification	Test Case TC-FR-47 (integration test)

ID	FR-48
Title	SMS Sent Only After Transfer Confirmation
Description	SMS notifications shall be sent only after a bank transfer is confirmed successful – not at the payment calculation or approval stage.
Priority	P2 – High
Source	Business Rule BR-25
Verification	Test Case TC-FR-48 (functional test)

ID	FR-49
Title	Multi-Language SMS
Description	The system shall support SMS message delivery in Sinhala, Tamil, and English. The farmer's preferred language shall be set on their profile and used for SMS content.
Priority	P3 – Medium
Source	Client Interview #3; Questionnaire
Verification	Test Case TC-FR-49 (localisation test)

4.12 Reporting

ID	FR-50
Title	Daily Collection Report
Description	The system shall generate a Daily Collection Report showing total kg per farmer and per route for a selected date.
Priority	P2 – High
Source	Client Interview #2; UC-18
Verification	Test Case TC-FR-50 (functional test)

ID	FR-51
Title	Monthly Payment Report
Description	The system shall generate a Monthly Payment Report showing gross payment, itemised deductions, and net payment per farmer for a selected month.
Priority	P2 – High
Source	Client Interview #2; UC-18
Verification	Test Case TC-FR-51 (functional test)

ID	FR-52
Title	Report Export
Description	All system reports shall be exportable in PDF and CSV formats.
Priority	P3 – Medium
Source	Client Interview #2
Verification	Test Case TC-FR-52 (functional test)

ID	FR-53
Title	Farmer Payment Receipt
Description	The system shall generate and allow printing of an individual payment receipt for each farmer upon successful payment processing.
Priority	P3 – Medium
Source	Client Interview #2
Verification	Test Case TC-FR-53 (functional test)

5. Non-Functional Requirements

5.1 Performance

ID	Requirement	Measure
NFR-01	The mobile app shall respond to QR scan and save a weight entry within 1 second under normal conditions.	< 1 second response time
NFR-02	The office desktop app shall load a monthly payment report for up to 500 farmers within 5 seconds.	< 5 seconds load time
NFR-03	A full-day route sync (up to 500 records) shall complete within 30 seconds on a 100 Mbps LAN.	< 30 seconds transfer
NFR-04	Monthly payment calculation for up to 1,000 farmers shall complete within 10 seconds.	< 10 seconds compute
NFR-05	The system shall support a minimum of 10 concurrent office desktop sessions without degradation.	10 concurrent users

5.2 Security

ID	Requirement	Verification
NFR-06	All network communication shall use TLS 1.2 or higher.	Security audit
NFR-07	All farmer personal and financial data shall be encrypted at rest using AES-256 or equivalent.	Security audit
NFR-08	Authentication tokens (JWT) shall be signed with RS256 and expire within the configured session timeout.	Security test
NFR-09	Bank API credentials and SMS API keys shall be stored in a secrets manager and never committed to source control.	Code review / inspection
NFR-10	The system shall maintain an immutable audit log of all actions affecting farmer profiles, deductions, and payment records.	Inspection
NFR-11	Approved payment records shall be cryptographically locked to detect unauthorised modification.	Security test

5.3 Usability

ID	Requirement	Verification
NFR-12	The mobile app shall use minimum touch targets of 48×48 dp (Android) / 44×44 pt (iOS) for all interactive elements.	UI inspection
NFR-13	A newly onboarded Collector shall be able to complete a full collection session with no assistance after a maximum of 30 minutes of training.	Usability test
NFR-14	The mobile app shall support display in Sinhala, Tamil, and English.	Localisation test
NFR-15	All error messages shall be written in plain language with clear corrective action guidance.	Heuristic review
NFR-16	The office desktop app shall follow standard Windows UI conventions (keyboard shortcuts, tab order, context menus).	Heuristic review

5.4 Reliability

ID	Requirement	Verification
NFR-17	The mobile app shall guarantee zero data loss for locally stored collection records, even on app crash or device shutdown.	Failure injection test
NFR-18	All database writes shall use ACID-compliant transactions to prevent partial data corruption.	Code inspection
NFR-19	Synchronisation operations shall be idempotent – repeating a sync shall not create duplicate records.	Idempotency test
NFR-20	Payment calculation results shall be deterministic – identical inputs shall always produce identical outputs.	Unit test
NFR-21	External API calls (bank, SMS) shall implement automatic retry with exponential back-off (max 3 retries).	Integration test

5.5 Scalability

ID	Requirement	Verification
NFR-22	The system shall support a minimum of 2,000 registered farmer records without performance degradation.	Load test
NFR-23	The system shall support a minimum of 5 concurrent collector devices synchronising simultaneously.	Load test
NFR-24	The backend API server architecture shall allow horizontal scaling (additional instances behind a load balancer) without code changes.	Architecture review

5.6 Maintainability

ID	Requirement	Verification
NFR-25	The codebase shall follow documented coding standards with inline comments for all business logic functions.	Code review
NFR-26	The system shall support database schema migrations with rollback capability using a migration framework.	Migration test
NFR-27	All configurable parameters (rate limits, session timeout, SMS templates) shall be manageable via admin UI without a code deployment.	Functional test
NFR-28	Unit test coverage for payment calculation and deduction logic shall be a minimum of 90%.	Coverage report
NFR-29	The system shall produce structured application logs in JSON format compatible with standard log aggregation tools.	Log inspection

6. Alternative Solutions & Feasibility Study

6.1 Alternative Solutions Analysed

Alternative 1 – Off-the-Shelf Agricultural Management Software

Several commercial agricultural management platforms (e.g., FarmERP, AgriSoft) are available that cover crop collection and payment features.

Aspect	Assessment
Description	Purchase or subscribe to an existing multi-purpose agricultural ERP system and configure it for tea collection.
Advantages	Rapid deployment. Vendor-supported. Known reliability. Reduced development risk.
Disadvantages	- Generic platforms lack Sri Lankan tea industry-specific workflows (e.g., route-based lorry collection, LKR payment integration, local bank API). - High licensing cost (USD 200–500/month). - Offline-first mobile collection is not a standard feature. - Customization is expensive and constrained by vendor APIs. - Sinhala/Tamil localization is typically not included.
Cost Estimate	LKR 600,000–1,200,000 per year (licensing). Additional customization: LKR 400,000–800,000 one-time.
Verdict	Not recommended. The system cannot be adequately configured for the specific operational model without prohibitive customisation cost.

Alternative 2 – Spreadsheet-Based Enhancement (Enhanced Legacy Approach)

Enhance the current process by replacing the legacy desktop application with a shared Microsoft Excel workbook (OneDrive or SharePoint) while keeping paper-based field collection.

Aspect	Assessment
Description	Office staff maintain a shared Excel workbook with payment formula templates. Collectors continue using paper route books; data is entered in the office.
Advantages	Very low cost (uses existing Microsoft 365 subscription). No development required. Immediately available.

Aspect	Assessment
Disadvantages	- Does not resolve the root cause pain points: manual transcription errors persist (P-01). - No mobile field data entry (P-01). - No QR identification (P-07). - No automated bank transfer (P-03). - No SMS notification (P-05). - Concurrent editing conflicts in shared spreadsheets. - No audit trail (P-04). - Not scalable beyond ~100 farmers without significant Excel complexity. - Poor data security.
Cost Estimate	- LKR 0 development cost. - Ongoing manual labor cost.
Verdict	Not recommended. Fails to address the core operational and accuracy problems identified in Section 3.2.2.

Alternative 3 – Recommended: Custom Multi-Platform System (TeaRoutePay)

Develop a purpose-built system comprising an offline-first mobile application, an office desktop application, and a backend API server tailored exactly to the tea collection business workflow.

Aspect	Assessment
Description	Custom cross-platform mobile app (Flutter), Windows desktop app (Flutter), and a Linux-hosted REST API backend.
Advantages	- Precisely addresses all seven pain points (P-01 to P-07). - Offline-first mobile design. - QR-based farmer identification. - Automated payment calculation and bank transfer. - SMS notification. Sinhala/Tamil localisation. - Full audit trail. - LKR fixed-point arithmetic.
Disadvantages	- Higher upfront development effort and cost compared to alternatives - Requires internal or contracted maintenance post-delivery.
Cost Estimate	- LKR 1,500,000–2,200,000 development (one-time). - LKR 120,000–180,000/year maintenance.
Verdict	Recommended. The only alternative that fully resolves all identified pain points within the operational and technical constraints.

6.2 Feasibility Study

6.2.1 Technical Feasibility

The system is designed as a microservices architecture handling financial transactions of approximately LKR 10 million per month. All required technologies are mature, well-documented, and appropriate for this scale:

1. **Cross-platform development (Flutter)** — A single Flutter codebase covers both the collector mobile app (Android, offline-first) and the office desktop application (Windows), reducing development effort while maintaining a consistent user experience across platforms.
2. **Backend microservices (Java Spring Boot)** — Spring Boot is the industry-standard framework for enterprise-grade, security-critical business APIs. Each microservice (Collection, Farmer, Payment, Deduction, Notification, Auth) is an independent Spring Boot application. Spring Boot's strong typing, transactional integrity via `@Transactional`, and mature ecosystem make it the correct choice for a system processing financial data at this scale.
3. **API Gateway and service discovery (Spring Cloud Gateway + Eureka)** — Spring Cloud Gateway handles all inbound routing, rate limiting, and authentication enforcement at the network edge. Eureka provides service discovery so microservices locate each other dynamically, consistent with the Spring Cloud ecosystem.
4. **Event streaming and audit trail (Apache Kafka)** — All financial events — payment calculations, deduction changes, transfer approvals — are published to Kafka topics. Kafka retains an immutable, ordered log of all events, providing a full audit trail for dispute resolution and compliance. This is a critical requirement for a system handling LKR 10 million per month.
5. **Mobile local storage (SQLite via Drift + Hive)** — SQLite managed through the Drift package provides typed, compile-safe offline SQL queries on the collector's device. Hive provides lightweight key-value storage for app state. Both support the offline-first collection workflow and pending-sync flagging.

6. **Cloud database (PostgreSQL + Flyway)** — PostgreSQL serves as the production-grade relational database, with each microservice maintaining its own schema. Flyway manages all schema migrations with version control, ensuring safe, repeatable database changes across environments.
7. **Caching and session management (Redis)** — Redis supports JWT token blacklisting (for secure logout), idempotency key storage (preventing duplicate payment processing on network retry), and inter-service rate limiting.
8. **Containerization and orchestration (Docker + Kubernetes on AWS EKS)** — Each microservice is containerized with Docker. Kubernetes manages deployment, scaling, and recovery in production. AWS Elastic Kubernetes Service (EKS) in the Singapore region (ap-southeast-1) provides the lowest latency to Sri Lanka with high availability guarantees.
9. **QR code generation and scanning** — ZXing handles server-side QR code generation for farmer identity cards. The mobile_scanner Flutter package handles scanning on the collector's device using the native camera API on Android.
10. **Security (Spring Security + JWT + RBAC + AES-256 + TLS)** — Spring Security provides enterprise-grade authentication and role-based access control across all services. JWT tokens with short expiry and Redis-backed blacklisting handle session management. AES-256 encrypts sensitive data at rest. TLS 1.3 encrypts all data in transit. AWS Secrets Manager stores all credentials and API keys — none are stored in source code or environment files. Spring Data Envers automatically maintains a tamper-evident audit log of every change to payment, deduction, and farmer records, recording the user, timestamp, and before/after values.
11. **Reporting (JasperReports)** — JasperReports is a Java-native reporting engine that integrates directly with Spring Boot, providing structured PDF generation for monthly payment reports, collection summaries, and deduction statements.

- 12. SMS notifications (Dialog Axiata SMS Gateway)** — Dialog Axiata provides a documented REST API for programmatic SMS delivery to Sri Lankan mobile numbers, used for payment notifications to farmers.
- 13. Bank transfer (LankaPay / Bank Transfer API)** — LankaPay is Sri Lanka's national payment infrastructure, providing a documented API for inter-bank transfers. This handles monthly payment disbursements directly to farmer bank accounts.
- 14. Monitoring and observability (Spring Actuator + Prometheus + Grafana)** — Spring Boot Actuator exposes health and metrics endpoints for all services. Prometheus collects metrics and Grafana provides real-time dashboards. Alerts are configured for payment service downtime and failed bank transfer events.

Conclusion: Technically feasible. All selected technologies are proven in production financial systems, actively maintained, and well-documented. The stack is internally consistent — Spring Cloud, Spring Security, Spring Data, and Spring Boot integrate natively, reducing integration risk. No unproven or experimental technologies are required.

6.2.2 Economic Feasibility

Cost Item	Estimate (LKR)	Notes
Software development	1,500,000 – 2,200,000	One-time, based on team size and timeline
Server hosting (cloud, 1 year)	60,000 – 120,000	Optional; not needed in LAN-only mode
SMS Gateway setup + first year	30,000 – 60,000	Depends on message volume
Bank API integration	0 – 50,000	Typically no fee; developer effort only
Annual maintenance	120,000 – 180,000	Post-delivery bug fixes and updates
Savings: staff time (manual re-entry)	~900,000/year	Estimated 3 staff hours/day × 300 days
Savings: payment error corrections	~200,000/year	Estimated dispute resolution time
Break-even point	~2.5 years	Development cost ÷ annual savings

Conclusion: Economically feasible. Break-even within approximately 2.5 years with ongoing annual savings thereafter.

6.2.3 Operational Feasibility

The system is designed for low-technical-proficiency users, ensuring that field and office operations can be executed with minimal training:

1. **Collector Mobile App:** Built with large touch targets, a localized Sinhala/Tamil user interface, and a 30-minute training target. Core tasks are streamlined into simple, offline-capable modules to minimize typing:
 - **Weight Collection:** A fast, 3-step flow (scan → weigh → save).
 - **Field Registration:** Guided, on-the-spot profile creation for new farmers directly from the field.
 - **Advances & Distribution:** Quick-select menus to instantly log financial deductions (e.g., cash advances) and record item distributions (e.g., fertilizer or tea packets) to a farmer's account.
2. **Owner / Office Staff Desktop Dashboard:** Adheres to standard Windows conventions with familiar tabular data entry. It automatically ingests complex field data—syncing newly registered farmers, updating inventory for distributed items, and applying field deductions to final payout calculations. It features a clean dashboard and a single-click approval flow, requiring no technical skills beyond basic current desktop usage.

Conclusion: Operationally feasible. The user interface design, automated offline syncing, and localized training plan specifically address the low-technical-proficiency user base, ensuring that advanced tasks like field registration and inventory distribution remain intuitive.

6.2.4 Schedule Feasibility

Phase	Duration	Key Deliverables
Phase 1 – Requirements & Design	8 weeks	This SRS, system architecture, UI wireframes, database schema
Phase 2 – Mobile App Development (Parallel)	8 weeks	Offline collection, QR scan, sync engine (Android + iOS)
Phase 3 – Desktop App Development (Parallel)	8 weeks	Farmer mgmt, deductions, payment calculation, reports
Phase 4 – Backend & Integrations (Parallel)	10 weeks	REST API, bank API, SMS gateway, cloud sync
Phase 5 – System Testing & UAT	4 weeks	Integration tests, performance tests, user acceptance
Phase 6 – Deployment & Training	2 weeks	Production deployment, staff training, pilot route
Total	24 weeks (6 months)	Full system go-live

6.3 Justified Recommendation

RECOMMENDATION: Alternative – Custom Multi-Platform System (TeaRoutePay) is the recommended solution. It is the only alternative that fully and directly addresses all seven identified pain points within the operational, technical, economic, and schedule constraints of the business. The higher upfront development cost is justified by a break-even point of approximately 2.5 years, elimination of manual transcription errors, accurate automated payments, real-time farmer notifications, and a full audit trail that protects the business from financial disputes.